

Wenyu Chiou

Ph.D. Candidate, Civil & Environmental Engineering, Lehigh University

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Education

Ph.D., Civil & Environmental Engineering, Lehigh University 2024 – present · expected 2028

Advisor: Dr. Y.C. Ethan Yang · CAWS Lab · Lehigh Center for Catastrophe Modeling and Resilience

M.S., Hydrological and Oceanic Sciences, National Central University Sep 2021 – Jun 2023

GPA 4.3/4.3

B.S., Earth Sciences, National Central University Sep 2017 – Jun 2021

GPA 3.9/4.3

Research Overview

My research asks a question generative AI has made urgent: when can a simulated human decision-maker be trusted? I approach it by owning every link of the evidence chain. I designed and fielded a 937-household flood-adaptation survey in New Jersey's Passaic River Basin, used structural equation modeling to identify how owners and renters actually decide among adaptation actions, and encoded those empirically identified pathways in a Bayesian-calibrated agent-based model of 52,141 households coupled to a catastrophe flood model with National Flood Insurance Program mechanics — including income-normalized equity analysis of who bears flood losses. With my advisor, I published a *Water Resources Research* article (62(6), e2025WR042111, 2026) substituting large language models as the behavioral engine of such simulations. That substitution creates a validity problem: nothing guarantees a generative agent decides the way real households do. My current work confronts it directly — testing whether LLM personas reproduce the decision pathways identified in my own survey data, and developing governance methods that validate agent behavior against physical constraints and behavioral theory — research software now in preparation for release. First-author manuscripts on the coupled model and on adaptation decision pathways are under review. The longer-term agenda is a validation standard for generative agents in consequential simulation.

Peer-Reviewed Publications

Yang, Y. C. E., & **Chiou, W.** (2026). Leveraging Large Language Models for Agent-Based Simulation of Human–Water System Interactions. *Water Resources Research*, 62(6), e2025WR042111.

<https://doi.org/10.1029/2025WR042111>

PUBLISHED

Manuscripts Under Review

First-author manuscript — coupled agent-based and catastrophe-model simulation of household flood adaptation (FLOODABM companion study).

UNDER REVIEW · *Journal of Hydrology*

First-author manuscript — flood-adaptation decision pathways across marginalized and non-marginalized groups, identified via structural equation modeling.

UNDER REVIEW · *Sustainable Cities and Society*

Conference Presentations

ISHC 2025, Tokyo, Japan — oral presentation 38-03, July 2025; co-authored with collaborators across Lehigh, FAU, Kyoto, and UTokyo.

AGU Fall Meeting 2025 — poster NH41E-0449 (Chiou, Yang, and Tanaka); Passaic River Basin case study.

AGU Fall Meeting 2023 — abstract; submarine groundwater discharge study combining electrical resistivity tomography (ERT), field observation, and numerical simulation.

Thesis

M.S. thesis, National Central University, 2023 — groundwater flow characteristics of the Taoyuan coastal zone.

Research Experience

Research Assistant, Surface Hydrology Group, Institute of Hydrological and Oceanic Sciences (IHOS), National Central University 2023 – 2024
Two- and three-dimensional groundwater modeling (HydroGeoChem); evaluation of nature-based solutions.

Summer Research Intern, National Science and Technology Center for Disaster Reduction (NCDR), Taiwan Summer 2022
IPCC AR6 downscaled climate ensembles; Python and MATLAB data pipelines.

Undergraduate Research Intern, Institute of Earth Sciences, Academia Sinica Jul – Aug 2020
Stable-isotope hydrology ($\delta D/\delta^{18}O$), Feitsui Reservoir.

Research Software

FLOODABM — 52,141-household coupled agent-based and catastrophe flood model of the Passaic River Basin, NJ, with National Flood Insurance Program mechanics, calibrated on a 937-household survey. Zenodo-archived companion code with CITATION.cff.

RESEARCH PROTOTYPE · archived companion code

research-hub-pipeline — local-first research-literature pipeline with a fail-closed anti-fabrication gate; v1.1.1 on PyPI; CI across 3 operating systems and 4 Python versions.

MAINTAINED OPEN-SOURCE TOOL

WAGF — research software in preparation for release.

Honors & Awards

Dean's Award, College of Earth Sciences, National Central University Jun 2023

NSTC Interdisciplinary Research Fellowship (Hydrology/Geophysics/Geology) 2021 – 2023

Best Undergraduate Student Project and Report, National Central University Nov 2020

NSTC Undergraduate Student Research Project Grant 2020 – 2021

Service

Graduate Student Committee, Department of Civil & Environmental Engineering, Lehigh University.